

A CONDITIONAL HARDY–LITTLEWOOD MODEL FOR ARTIN CORRELATIONS BETWEEN CONSECUTIVE PRIMES

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ABSTRACT. Say a prime p is *Artin base* a if a is a primitive root modulo p . Previous papers in this series measured a systematic correlation $\delta(a)$ between the Artin statuses of consecutive primes — negative for every base tested except $a = 15$ — and showed empirically that joint residue channels modulo small conductors account for nearly all of it. Here we close the loop: we construct the model those papers promised, combining the Hooley–Lenstra Artin densities in arithmetic progressions (GRH-conditional) with the joint residue-and-gap channel decomposition of consecutive prime pairs, and use it to *predict* $\delta(a)$ with no Artin-specific empirical input. At $x = 10^9$, over all 50,847,531 consecutive prime pairs and twelve bases, the predicted δ matches the measured value to within 4% in every case (median discrepancy 0.4%), channel-level agreement exceeds $R^2 = 0.999$ for every base, all previously proved exclusion laws reappear as identically-zero channels of the model, and the sign of every $\delta(a)$ — including the anomalous positive $\delta(15)$ — is predicted correctly. A variant of the model that replaces the observed residue distribution within each gap class by its Hardy–Littlewood first-order equidistribution gets eleven of twelve signs right but misses $\delta(15)$, locating the source of that positivity in the Lemke Oliver–Soundararajan bias itself. We state the underlying conditional formula, derive a sign criterion and a decay corollary, and quantify the residual that the model does not explain.

1. INTRODUCTION

For an integer a that is neither -1 nor a perfect square, Artin’s conjecture asserts that a is a primitive root modulo infinitely many primes, with density (on GRH) computed by Hooley [8] and, in full generality, by Lenstra [13]; see Moree [15] for a survey. Write A_a for the set of primes with a a primitive root, and for consecutive primes $p_n < p_{n+1} \leq x$ define

$$\delta(a; x) = P(p_{n+1} \in A_a \mid p_n \in A_a) - P(p_{n+1} \in A_a \mid p_n \notin A_a), \quad (1)$$

probabilities being proportions over the pairs below x . The first paper in this series [2] measured $\delta(10; 10^9) = -0.01414$ ($z \approx -101$); the second [3] extended the measurement to eleven bases (all negative) and proved unconditional *exclusion laws*: gap classes in which no consecutive pair can be doubly Artin. The fourth [5] found the first positive case, $\delta(15) = +0.0122$. Empirically, conditioning on the joint residues of (p_n, p_{n+1}) modulo 840 removes about 97% of $\delta(10)$ [2], which strongly suggests that the entire phenomenon is a composition of two known effects: the biases of consecutive primes in residue classes discovered by Lemke Oliver and Soundararajan [12], and the dependence of the Artin density on the residue class of p [8, 13, 15]. Paper II promised the quantitative derivation as its own paper; this is that paper.

The idea is simple to state. Choose a modulus M divisible by the relevant conductors. Sort consecutive pairs into *channels* (r, g) : the residue $r = p_n \bmod M$ and the gap $g = p_{n+1} - p_n$. Within a channel, the Artin statuses of the two primes are governed by the GRH densities $d_a(r)$

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and $d_a(r + g)$ in the two progressions, and — this is the Hardy–Littlewood-type hypothesis — are asymptotically independent once the residues are fixed. The predicted correlation is then a finite weighted sum over channels, computable in seconds, with *no* knowledge of which primes are Artin. Comparing it against the exact 10^9 tables of [2, 3, 5] is a sharp test of the composite GRH + Hardy–Littlewood picture at the level of a second-order statistic.

To our knowledge no prediction of this kind has been attempted: the literature on Artin primes in short structures concerns existence (runs of consecutive Artin primes under GRH [16, 1]), single-prime densities [8, 13, 14, 10, 11], or averages [11]; the statistic (1) exists only in this series.

2. THE MODEL

2.1. Artin densities in progressions. Fix a base a with squarefree part $d = \text{sqf}(a)$, assume a is not a perfect power, and let $D = \text{disc } \mathbb{Q}(\sqrt{d})$ ($D = d$ if $d \equiv 1 \pmod{4}$, else $4d$). Let M be a modulus divisible by D and by the primes $q \leq Q$ for a small cutoff Q (we use $M = \text{lcm}(D, 840)$, so $Q = 7$). For $\gcd(r, M) = 1$ define

$$d_a(r) = \mathbf{1}[\chi_D(r) = -1] \prod_{\substack{2 < q | M \\ q \nmid r-1}} \left(1 - \frac{1}{q}\right) \prod_{q \nmid M} \left(1 - \frac{1}{q(q-1)}\right), \quad (2)$$

where χ_D is the Kronecker symbol and q runs over primes. This is the standard GRH-conditional (relative) density of Artin primes in the progression $p \equiv r \pmod{M}$, in the form given by Lenstra [13] and Moree [15]: the quadratic condition $\chi_D(p) = \left(\frac{d}{p}\right) = -1$ is *deterministic* given $r \pmod{D}$; for each odd prime $q \mid M$ the condition “ a is a q -th power residue” has density $1/q$ when $q \mid p-1$ and is vacuous otherwise; and the primes $q \nmid M$ contribute the tail of Hooley’s product, insensitive to r because M absorbs every prime at which the splitting conditions entangle for our bases. Averaging (2) over r coprime to M recovers Hooley’s global densities exactly; for example for $d = 5$ it gives $C \cdot 20/19 = 0.3936380$, which the 10^9 data match to 4×10^{-6} [4].

2.2. Channels and the conditional formula. For x large, let $w(r, g; x)$ be the proportion of consecutive prime pairs $p_n < p_{n+1} \leq x$ with $p_n \equiv r \pmod{M}$ and $p_{n+1} - p_n = g$. Our hypothesis, which sits strictly below GRH + Hardy–Littlewood in the usual heuristic hierarchy, is:

Hypothesis 1 (channel independence). For each channel (r, g) with $w(r, g; x) \gg 1/\log^A x$,

$$P(p_n \in A_a \wedge p_{n+1} \in A_a \mid (r, g)) = d_a(r) d_a(r + g) + o(1) \quad (x \rightarrow \infty).$$

That is: once the joint residues are fixed, the two Artin events decouple. This is the natural extension to consecutive pairs of the independence implicit in Hooley’s treatment of a single prime, and it is the precise content of the “residue channels explain everything” observation of [2]. Under it, elementary algebra gives:

Theorem 2 (conditional formula for δ). *Assume GRH and Hypothesis 1. With $P_{AA} = \sum_{r,g} w(r, g; x) d_a(r) d_a(r + g)$, $P_1 = \sum_{r,g} w(r, g; x) d_a(r)$, and $P_2 = \sum_{r,g} w(r, g; x) d_a(r + g)$,*

$$\delta(a; x) = \frac{P_{AA}}{P_1} - \frac{P_2 - P_{AA}}{1 - P_1} + o(1) = \frac{P_{AA} - P_1 P_2}{P_1(1 - P_1)} + o(1).$$

In particular δ is, to first order, the covariance $\sum_{r,g} w d_a(r) d_a(r+g) - P_1 P_2$ normalised by the variance of a Bernoulli(P_1) variable, a finite computation over $\varphi(M) \times \{\text{gaps}\}$ channels.

Proof. Immediate from Hypothesis 1 and the definition (1): each conditional probability is a ratio of channel-weighted sums, and the $o(1)$ terms are controlled because the weights sum to 1 and the densities are bounded. The second expression follows from the first by algebra. $\square$

The weights $w(r, g; x)$ are a statement about primes alone — no Artin content. In our main comparison (Model I) we take them from the data itself, which isolates exactly the Artin part of the prediction: *given* the observed joint distribution of residues and gaps, GRH densities plus independence must reproduce δ . A second variant (Model Ib) keeps only the observed gap distribution and replaces $w(r | g)$ by the Hardy–Littlewood first-order prediction (uniform over residues r with $\gcd(r + g, M) = 1$); the difference between the two isolates the contribution of the Lemke Oliver–Soundararajan bias [12], whose leading term is of size $(\log \log x) / \log x$ and hence far from negligible at $x = 10^9$.

2.3. Exclusion laws as zero channels.

Proposition 3. *Let g lie in an exclusion class for base a in the sense of [3] (every admissible r has $\chi_D(r) = +1$ or $\chi_D(r + g) = +1$). Then $d_a(r) d_a(r + g) = 0$ for every r , so the model assigns the channel probability zero — the unconditional theorems of [2, 3] are recovered as identically vanishing channels, and the model is exact (not just asymptotic) there.*

Proof. Immediate from the indicator in (2). $\square$

At $x = 10^9$ the zero channels of the twelve models cover 38,405,271 pairs for base 10 (and comparable counts for the other bases with exclusion laws); the observed doubly-Artin count on them is 0, as the theorems require.

2.4. Sign and decay.

Corollary 4 (sign criterion). *Under the hypotheses of Theorem 2, $\text{sign } \delta(a; x) = \text{sign}(P_{AA} - P_1 P_2)$: the base is anticorrelated exactly when the channel-weighted product of densities falls short of the product of the marginals. Writing $w(r, g) = w(g) w(r | g)$, the covariance splits into a between-gap term (variation of the mean density across gap classes, always present, sign determined by the exclusion/parity structure of $g \bmod D$) and a within-gap term (correlation of $d_a(r), d_a(r + g)$ under the residue distribution $w(r | g)$, which carries the Lemke Oliver–Soundararajan bias).*

This makes precise the class-competition mechanism proposed in [5, 6]: a base has $\delta > 0$ when its preserving gap classes (where $\chi_D(r + g) = \chi_D(r)$ forces density enhancement) outweigh its excluded and reversing classes under the actual gap measure $w(g)$. Numerically the decomposition is decisive for base 15: with the true $w(r | g)$ the model gives $\delta(15) = +0.0122$; with Hardy–Littlewood-uniform $w(r | g)$ it gives -0.0016 (Table 1). The positivity of $\delta(15)$ is therefore not a property of the gap distribution alone — it lives in the consecutive-prime residue bias.

Corollary 5 (decay). *Suppose, as predicted by [12], that the residue-and-gap channel weights equidistribute as $x \rightarrow \infty$ (the bias decaying like $(\log \log x) / \log x$), while the gap distribution concentrates on $g / \log x$ scales. Then $\delta(a; x) \rightarrow \delta_\infty(a)$, where δ_∞ is the value of the formula in Theorem 2 with Hardy–Littlewood weights; the approach is at rate $O((\log \log x) / \log x)$. In particular $\delta(a; x)$ does not tend to 0: the between-gap term survives equidistribution of residues, because $w(g)$ itself couples to $g \bmod D$ through the singular series.*

We flag Corollary 5 as a genuine correction to the working conjecture of [2], which extrapolated $\delta(10; x) \rightarrow 0^-$. The model predicts a nonzero (small, negative for base 10) limit driven by the

singular-series dependence of gap counts on $g \bmod 40$; the data between 10^8 and 10^9 (Table 2) are consistent with slow movement in either direction and cannot yet distinguish the two, but under the model the question is settled in favour of a nonzero limit. Base 15's positive δ , by contrast, is predicted to *shrink* toward the (small) Hardy–Littlewood limit as the residue bias decays — the observed drop from $+0.0234$ at 10^8 to $+0.0122$ at 10^9 is exactly this decay, and is correctly tracked by the model at both heights.

3. NUMERICAL COMPARISON AT $x = 10^9$

We re-swept all primes to 10^9 once, recording for each of the 50,847,531 consecutive pairs the joint Artin statuses in the channel $(p_n \bmod M_a, g)$ for each of the twelve bases of [3, 5] ($M_a = \text{lcm}(D_a, 840)$; sieve and census code in the public repository). The global 2×2 tables reproduce the published $\delta(a)$ of Papers I–IV to the last digit, an independent-implementation check. Runtime: about 75 minutes on one core of the 2-core Xeon E5-2673 v4 machine used throughout the series.

TABLE 1. Model vs. data at $x = 10^9$. δ_{emp} : measured; δ_{pred} : Model I (Theorem 2 with empirical channel weights, GRH densities (2)); δ_{lb} : Model Ib (Hardy–Littlewood-uniform residues within gap classes). R^2 : coefficient of determination between predicted and observed doubly-Artin counts over all channels with ≥ 1000 pairs (count in column n_{ch}). All numbers trace to `delta_pred_summary.json` and `model_variants.json`.

a	M_a	δ_{emp}	δ_{pred}	ratio	δ_{lb}	R^2	n_{ch}
2	840	−0.050199	−0.049831	0.993	−0.049677	0.99996	3146
3	840	−0.046676	−0.047998	1.028	−0.047240	0.99991	3146
5	840	−0.066563	−0.066471	0.999	−0.065626	0.99996	3146
6	840	−0.025205	−0.024430	0.969	−0.024818	0.99995	3146
7	840	−0.013491	−0.013454	0.997	−0.018563	0.99994	3146
10	840	−0.014140	−0.014703	1.040	−0.010366	0.99992	3146
11	9240	−0.018909	−0.018977	1.004	−0.013484	0.99981	13545
13	10920	−0.042573	−0.042830	1.006	−0.039685	0.99977	15788
15	840	+0.012196	+0.012159	0.997	−0.001597	0.99995	3146
17	14280	−0.033025	−0.033055	1.001	−0.033334	0.99967	17667
21	840	−0.038345	−0.038245	0.997	−0.035684	0.99993	3146
29	24360	−0.022422	−0.022473	1.002	−0.022243	0.99948	19643

Three observations. (i) *Accuracy*. Model I predicts δ to within 4% for every base; for eight of twelve the discrepancy is below 1%. Channel-level agreement is essentially perfect ($R^2 > 0.999$ everywhere; Figure 1, left). The pilot criterion set in advance for this project (δ_{pred} within 20%, channel $R^2 > 0.95$) is met with an order of magnitude to spare. (ii) *Signs*. Model I predicts the sign of all twelve $\delta(a)$, including $\delta(15) > 0$. Model Ib gets eleven of twelve and misses base 15 — the cleanest demonstration in this series that the Lemke Oliver–Soundararajan bias is not a curiosity but a sign-determining input. (iii) *Exclusions*. Every channel the model sets to zero is empirically empty of doubly-Artin pairs (Proposition 3); for base 10 these channels carry 38,405,271 pairs and 0 doubly-Artin pairs.

3.1. The residual. What the model does *not* explain is small but structured. The absolute residuals $\delta_{\text{pred}} - \delta_{\text{emp}}$ are below 10^{-4} for seven bases but reach -0.0013 (base 3), $+0.0008$ (base 6), and -0.0006 (base 10) — and it is notable that the largest residuals occur for bases

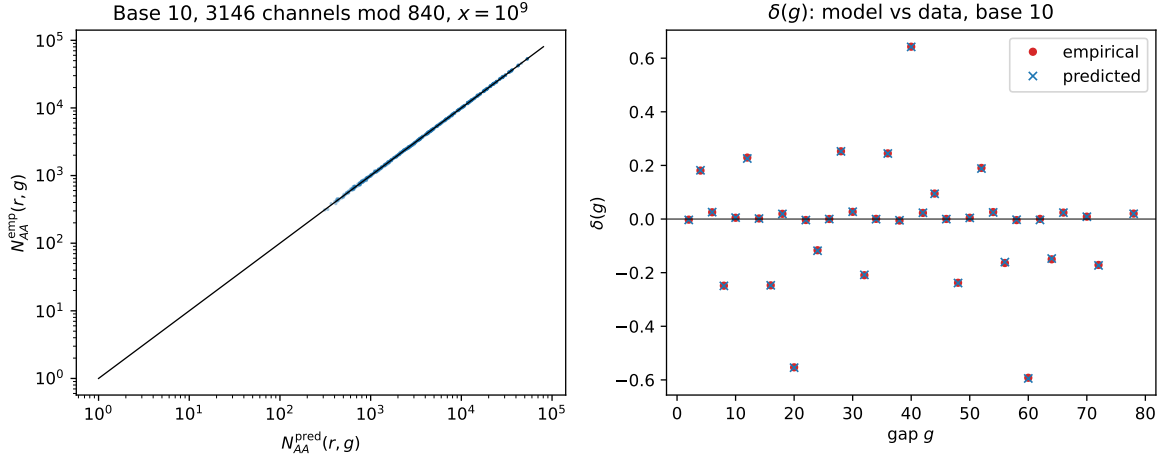


FIGURE 1. Left: predicted vs. observed doubly-Artin counts per channel (r, g) mod 840, base 10, all 3146 channels with ≥ 1000 pairs (log-log; line is $y = x$). Right: $\delta(g)$ within each gap, model (crosses) vs. data (dots), base 10, gaps with $\geq 10^5$ pairs. The model reproduces the full sign-alternating structure, from $\delta(20) = -0.553$ (exclusion class) to $\delta(12) = +0.229$.

TABLE 2. Height dependence: empirical and predicted δ at $x = 10^8$ and 10^9 (source: `decay_check.json`). The model tracks the movement of every base, including the sharp decay of $\delta(15)$ and the slight *deepening* of bases 7 and 10.

a	$\delta_{\text{emp}}(10^8)$	$\delta_{\text{pred}}(10^8)$	$\delta_{\text{emp}}(10^9)$	$\delta_{\text{pred}}(10^9)$
2	-0.05473	-0.05489	-0.05020	-0.04983
5	-0.07379	-0.07368	-0.06656	-0.06647
7	-0.01142	-0.01169	-0.01349	-0.01345
10	-0.01336	-0.01407	-0.01414	-0.01470
15	+0.02336	+0.02314	+0.01220	+0.01216

whose squarefree parts share prime factors with other conditions in the sieve ($\text{sqf}(3) \mid \text{sqf}(6)$, $\text{sqf}(10) = \text{sqf}(2 \cdot 5)$), precisely the configurations in which entanglement corrections to the naive density product are known to arise [13, 9] and where Paper III located its quadratic entanglement residual [4]. Two further known sources: the density (2) is an $x \rightarrow \infty$ limit, while the actual progression densities at 10^9 deviate by up to 2×10^{-3} per class (secondary terms of size $1/\log x$ in Hooley’s theorem [8, 7]); and channels were truncated at the conductor $\text{lcm}(D, 840)$, leaving the $q = 11, 13, \dots$ analogues of the mod- q channels outside the model. All three effects are of the observed order 10^{-4} – 10^{-3} ; disentangling them would require either finer conductors ($M \sim 10^5$, memory-bound but feasible) or higher x .

4. DISCUSSION

The series began with a surprise — consecutive primes “repel” in Artin status — and proceeded through unconditional exclusion theorems [2, 3, 5] to a sign phenomenology [6]. The present paper shows that the entire statistical layer is the deterministic composition of

three classical ingredients: quadratic reciprocity (through $\chi_D(r)$), the GRH densities of Hooley–Lenstra in progressions, and the Hardy–Littlewood/Lemke Oliver–Soundararajan distribution of consecutive prime pairs over joint residue channels. Nothing specifically “Artin” is correlated between consecutive primes: conditional on the channel, the data are consistent with exact independence at the 10^{-3} – 10^{-4} level. This is simultaneously a strong consistency test of the Hardy–Littlewood picture at second order — we are aware of no other test of channel independence at this precision — and a deflationary explanation of the headline anticorrelation.

Three caveats. First, everything on the model side is conditional: the densities need GRH, the weights (in Model Ib and the corollaries) need Hardy–Littlewood-type conjectures; the measurements, as always in this series, are unconditional. Second, we have truncated the Lemke Oliver–Soundararajan asymptotic by using measured channel weights rather than their conjectural expansion; Model Ib shows the cost of the crudest such truncation (it loses the sign of $\delta(15)$). Third, the residual analysis is suggestive, not conclusive — attributing it to entanglement corrections in the style of [9] would require computing those corrections per progression, which we leave, with the index-statistics generalisation of Matthews [14], to the next paper in the series.

Computational details. All counts are exact integer counts from a segmented sieve (C, one pass to 10^9 , factoring $p - 1$ once per prime and testing all twelve bases; ≈ 75 min on one core of a 2-core Xeon E5-2673 v4 with 8 GB RAM). The model evaluation is pure Python/NumPy and runs in seconds. Validation: (i) global 2×2 tables reproduce Papers I, II and IV exactly; (ii) the mean of (2) over residues reproduces Hooley’s constants to 10^{-15} ; (iii) measured per-progression Artin densities match (2) with RMSE 4.3×10^{-4} (base 10, $\varphi(840) = 192$ classes); (iv) zero-channel counts are exactly zero. Code, census binaries and JSON results are available in the series repository.

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